USER MANUAL

RTM

The toolkit that tells you whether the new master is actually better.

RELEASE

RTM v5.0.5
macOS arm64 · Windows x64

SIGNED · NOTARIZED

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DOCUMENT

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The toolkit that tells you whether the new master is actually better, which platforms will hate it, and exactly how to fix it before delivery.

v5.0.5 · macOS arm64 + Windows x64 · everything runs locally

What's in the box

Three apps. They cover the full mix → master → deliver loop, and they all talk to each other over a local handoff folder. No accounts, no cloud, no upload.

APP	WHAT IT DOES	WHO REACHES FOR IT
RTMCOMPARE	A/B compare, single-file QC, album batch, Atmos analysis, streaming-platform preview, master-chain render	Mixing + mastering + label QC
RTMPROFILE	Drop a corpus of your masters → out comes a `.json` profile. Loads into RTMcompare's Match tab as a "sound like Ohad" target curve	Mastering engineers building a personal target reference
RTM SEND	VST3/AU/Standalone plugin. One button on your DAW master bus → audio + DAW context lands in RTMcompare	Mixing engineers, producers, anyone who wants to skip the bounce-and-drag dance

The whole stack is signed (Apple Developer ID), notarized, stapled. First-launch is clean — no Gatekeeper warning, no "this app is from an unidentified developer" workaround.

Install

macOS (arm64 / Apple Silicon)

1. Download the DMG.
2. Drag the app to **/Applications**.
3. Eject the DMG.
4. Launch from /Applications. (If you launch from inside the DMG window macOS App Translocation runs the app from a sandbox that breaks our bundled Python — the app refuses to start and tells you to drag it to /Applications first.)

Three DMGs:

- **RTMcompare-5.0.5-arm64.dmg** — the analyzer
- **RTMprofile-1.0.0-arm64.dmg** — the profile builder
- **RTM-Send-1.0.0.dmg** — drag the AU/VST3/Standalone bundles into their folders, or double-click `Install RTM Send.command` to do it for you

Windows (x64)

- **RTMcompare Setup 5.0.5.exe** — NSIS installer (signed)
- **RTMcompare 5.0.5.exe** — portable build (no install, runs in place)
- **RTMprofile Setup 1.0.0.exe** + **RTMprofile 1.0.0.exe** — same pattern

The Windows plugin (.vst3) is built separately on a Windows host via GitHub Actions. Drop the resulting installer into Common Files\VST3.

The five workflows (read this section)

1. Two-file compare — the main event

Drop **A** (the reference, the rough mix, the previous master) and **B** (the new master, the candidate). Hit **Compare**.

You get seven tabs:

- **Overview** — LUFS / TP / LRA / dynamics deltas, level-matched waveforms, level-matched A/B player. The headline numbers, in one place.
- **Mastering** — the move your master pass actually made: per-band shape change, transient density, limiter aggressiveness, stereo width per band. (See *Mastering Delta* below.)
- **Stereo & Spectrum** — vectorscope timeline, phase correlation per band, mono-fold-down per band, masking overlap.
- **Quality** — clip / hum / click / distortion / generation-loss detection. Catches AAC-of-AAC-of-WAV laundered files.
- **Delivery** — Streaming Preview table for every major platform, Apple Digital Masters check, BWF metadata pane, Spec Drift badge.
- **Atmos** — appears when you drop an ADM BWF; channel layout, binaural TP, downmix vectorscope, downmix phase timeline.
- **Match** — the target-curve panel. Pick a reference profile (an engineer's, an album average, your own RTMprofile output), see the per-band correction, export it as a real DAW preset.

Tabs are sticky to the top of the scroll. You can bury yourself in a panel and still see what tab you're on.

2. Single-file QC

One file, no reference. Click **Analyze Reference Only**.

Use this when a master lands on your desk and you need a clean delivery readout — LUFS, TP, LRA, streaming previews, clip detection, spec compliance — without picking a reference. Faster than a full compare; produces the same Delivery tab.

3. Album batch

Drop a folder of WAVs / AIFFs / FLACs. Every track gets scanned for LUFS, TP, LRA, ISRC, sample-rate consistency. Outliers vs the album median get flagged in red.

Inside the batch:

- **Cohort Mode** — pin one track as the cohort reference. Every other track gets a heatmap distance score across 31 bands plus an RMS distance column. Sort by distance, see the one track that strays furthest from the family resemblance.
- **Loudness anchor** — flip the Δ column from *vs album median* to *vs Spotify −14*, *Apple −16*, *R128 −23*. Read delivery headroom directly off the table.
- **Reissue mode** — pin "old master" as the A across every song tab for reissue workflows. A/B every track of the reissue against its original.
- **Per-song notes + album notes** — both ride along in the PDF export.

The right-hand song slot rotates as you click rows in the table, so you don't end up with thirty tabs open.

4. Atmos / ADM

Drop a multichannel ADM BWF. The analyzer auto-routes to the Atmos surface: channel layout, programme name, binaural true-peak, downmix vectorscope, downmix phase correlation timeline. The A/B player auto-loads the Atmos stereo downmix against your stereo reference, level-matched.

Atmos Solo mode handles single-file Atmos QC (no reference).

5. Plugin → app handoff

Install **RTM Send** in your DAW. Drop it on the master bus. Hit the **Send** button.

The plugin writes the rendered audio + DAW context (region name, session name, sample rate, BPM, key, region time range) into `~/.rtm/incoming/`. RTMcompare's file watcher picks it up within ~1 second and routes it:

- **Single** → loads as File A, runs analyser-only
- **Compare B** → loads as File B in the active compare

- **Batch** → seeds File A; click Analyze Album to start a batch with this track as track 1

A chip appears in the banner showing the DAW context and any region timestamp. No bounce dialog, no drag-drop.

RTMprofile — building your own target reference

Open RTMprofile. Drop your back catalogue (5+ tracks recommended for a stable curve). Type your name + role + genres. Click **Build profile**.

The Python side (shared with RTMcompare's bundle) measures, per file:

- LUFS-I via pyloudnorm
- True peak (4× resampled per BS.1770-4)
- LRA, peak-to-RMS, crest
- Third-octave spectrum (31 bands, Welch's method, mean-centred)
- Stereo width

Then it aggregates: median curve, mean+std for the scalars. Out comes a `.json` profile that drops into `~/.rtm/profiles/<slug>.json` and auto-appears in RTMcompare's Match tab.

The output is a **target reference** for your style. Match against it in any future session and the Master Assistant computes the per-band correction needed to land in your sonic neighbourhood. Useful for:

- Building a *house style* for a label's roster
- A mastering engineer maintaining consistency across years of work
- A producer cloning their favourite-sounding records into a personal target curve

Privacy: all measurement runs locally. The profile JSON contains anonymised numbers — no audio, no filenames in the output (only the sample count).

RTM Send — the plugin

JUCE-based. Ships as VST3 + AU + Standalone on macOS, VST3 + Standalone on Windows (AAX is on the roadmap once we have a PACE Eden cert).

Three capture modes:

- **Last-N seconds** — a lock-free SPSC ring buffer captures the most recent 30 s (configurable up to 120 s) of audio at the master bus. Default mode. Always available.
- **Loop region** — uses your DAW's loop points. Plays the loop once through, captures it, ships it. Greys out gracefully on hosts that don't expose loop info via the standard playback callback.
- **Triggered region** — manual Rec / Stop. Host-agnostic. Works in any DAW, including those that don't expose loop points.

ARA2 (regions catalogue, "send Track 03 of the Wavelab montage without master-bus playback") is wired up but currently disabled in the shipped binary — it lights up on the next plugin release.

Three send routes:

- **Single** — the plugin writes a `.wav` + `.rtm.json` sidecar + `.ready` marker into `~/.rtm/incoming/`. RTMcompare loads it as the current single-file analysis.
- **Compare (File B)** — same write, different sidecar route hint. Slots into your active two-file compare as B.
- **Album batch** — seeds the batch with this track as track 1.

Rendered audio uses the host's sample rate; sidecar carries the DAW context (session name, region name, ARA region bounds if relevant) plus a `pluginVersion` stamp so RTMcompare knows what wrote it.

Why three apps and not one DAW plugin that does everything? Because the analysis is heavy (Demucs stem separation, 4× true-peak, AAC encoding for the streaming preview, third-octave spectrum across 31 bands, the whole Mastering Delta panel) and an audio plugin's host process is the worst possible place to do that work. The plugin is a 1 MB capture-and-handoff agent. The 256 MB analyser sits in its own process.

The headline panels

Mastering Delta

The seventh tab on every two-file compare. Treats the comparison as *"what did this master pass actually change?"* and gives you a signed report card:

- Broadband loudness gain (signed, in dB)
- Per-band tonal shape across 31 1/3-octave bands, mean-centered so it's the *shape* move, not the *level* move
- LRA delta · PSR delta · RMS-to-peak delta
- Limiter aggressiveness estimate (TP delta-derived)
- Transient density change (%)
- Stereo width change per band
- TP overs pulled back (count)
- **Playback delta after platform normalization** — what the listener hears once Spotify / Apple / etc. re-equalize the loudness. On hot masters, often reads 0.0 dB on every platform — that's the point: streaming normalization wipes the mastering loudness gain. Useful insight, not a bug.
- **Mastering signature hash** — 8-character fingerprint of the rounded per-band shape. Same hash twice = identical move. Useful for archival and version-tracking.

Sound Check twin

Teal-green ≈ button next to each platform row in the Streaming Preview. Renders 30 s of the loudest section of your track *through that platform's actual ingest chain*: normalization gain → 4× oversampled true-peak limiter modeled on Apple's, with the per-block gain-reduction envelope visible in the Streaming Delta Heatmap → AAC 256 kbps via Apple's `afconvert` on macOS (ffmpeg fallback on Windows).

Hit ≈, listen, decide. The fastest answer to *"will my master clip on Spotify"* you'll ever get.

Streaming Preview

A live table showing what each platform will *actually play your master at*: normalization action, played LUFS, played TP, breach flags. Spotify / Spotify Loud / Apple Music / YouTube / Tidal / Amazon / Deezer / SoundCloud, plus short-form (TikTok / Reels / Shorts).

Every spec is pinned in a versioned registry — published date, revision date, source citations. Reload an old result a year from now and a **Spec Drift** badge tells you whether what you measured against is still current.

Match tab + Master Assistant

Pick a reference profile (an engineer's, an album average, an RTMprofile output, or a streaming target). The Master Assistant computes a per-band corrective EQ, optional dynamics, optional limiter — and lets you:

- **Audition it live** through the A/B player with an Amount fader (0-100%) so you can hear the move at any intensity
- **Export the EQ** as a real DAW preset:
- **FabFilter Pro-Q 3 / 4** — native binary `.ffp` (Pro-Q 4 reads Pro-Q 3 binaries unchanged)
- **FabFilter Pro-Q text** — paste into Pro-Q's Paste bar
- **Ableton EQ Eight** — `.adv` (gzipped XML, real Live 12 schema)
- **CSV / JSON / clipboard** — for everything else
- **Apply and bounce** — render a 24-bit WAV with the EQ baked in, optional true-peak limiter at −0.3 dBTP (Apple Music spec, the strictest common ceiling), BWF chunk pre-stamped

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Translation Check

Four playback environments rendered as 30 s `.m4a` auditions you can A/B against the original:

- **Phone speaker** — modern phone driver; sub disappears, presence range dominates
- **Earbuds** — consumer earbuds (AirPods-ish)
- **Club PA** — house-system PA with mono-sum sub
- **Car cabin** — generic mid-class consumer car cabin

The renderer tells you the lost-LF energy and the presence-band change so you can read what you're about to hear before you hit play.

Per-stem AI detection (Deep Scan)

Demucs separation: vocals / drums / bass / other. Each stem gets its own AI-content verdict. Tells you whether the AI suspicion is concentrated in one stem (often: vocals) or spread broadly. A vocal-cloned-but-instrumentation-organic mix scores very differently from a fully synthetic one.

Reference Library

Star a track in the recent-references row. It persists across restarts. The library shows quick-scan stats per reference (LUFS, TP, BPM, key) so you can pick the right reference in three seconds.

Working modes

The header has a **Surface picker** with five modes that change which panels appear by default:

- **Music** — Spotify / Apple Music / YouTube / Tidal / Amazon at top. Broadcast hidden. Atmos hidden.
- **Full** — everything on. Music + broadcast + Atmos.
- **Bcast** — broadcast first: R128 / ATSC A/85 at top, dialog gate prominent.
- **Netflix** — Netflix Sound Mix Specifications v1.6 (−27 LKFS dialog anchor, −2 dBTP ceiling, 5.1 + stereo, 48k/24-bit minimum).
- **Post** — Atmos / immersive. ADM validation surfaced.

Two side toggles:

- **Advanced QC** — reveals collapsed-by-default diagnostic panels (Mono Compatibility, Phase Bands, Transient Density, Tempo Drift, Waveform Diff, Masking Overlap). Off by default because most engineers don't need them most of the time. Needs Deep Scan to populate.
- **Educator** — adds a *why this matters* explainer to every panel. Useful when handing the report to a producer or onboarding a junior engineer.

Keyboard shortcuts

Press `?` anywhere to see the full list. The ones you'll use:

KEY	ACTION
SPACE	A/B player play/pause
TAB	toggle A / B
M	mono fold-down on the player
1 – 7	jump to tabs 1–7 in compare view
[/]	set loop start / loop end
\	clear loop
← / →	step between songs in batch song-tab
⌘/CTRL + K	command palette — search any metric, jump to its tab
⌘/CTRL + E	export EQ
⌘/CTRL + ⇧ E	apply EQ + bounce

Where your data lives

Everything stays on your disk:

- **macOS**: `~/Library/Application Support/RTMcompare/python-cache/` — Python bytecode + Numba JIT cache, redirected here so the signed app bundle is never written to (codesign integrity stays intact)
- `~/.rtm/` — analysis history, references library, plugin inbox, custom DSP profiles. Lives in your home folder so it survives app reinstalls
- `~/.rtm/profiles/` — engineer profile JSONs (RTMprofile output lands here)

No telemetry. No phone-home. No analytics.

The only network calls happen when you sign in with a license — and even then the app works fully offline for 180 days after first activation.

Troubleshooting

"RTMcompare cannot be opened from inside the DMG" You launched from the DMG window. macOS App Translocation runs DMG-launched apps in a sandbox that can't reach our bundled Python. Drag to /Applications, eject the DMG, open from /Applications.

"Preparing audio…" stuck on the player Pre-5.0 bug. Update to 5.0.5+.

Sound Check twin shows "render ×" on Windows Install ffmpeg (`winget install ffmpeg`) and add it to PATH. Restart.

Advanced QC panels are blank You're viewing a Fast scan result. Re-run with Deep Scan.

Mastering Delta panel says 0.0 dB on every platform Working as designed — both files are louder than every platform's normalization target, so streaming normalization attenuates them to the same level. The metric is correctly telling you that mastering loudness gains evaporate after platform normalization.

Plugin → App handoff doesn't show up in RTMcompare Check `~/.rtm/incoming/` exists and the plugin can write to it. RTMcompare must be running for the file watcher to fire. The plugin writes `.ready` last; if you see `.wav` + `.rtm.json` but no `.ready` , the plugin crashed mid-write — check the host's plugin log.

RTMprofile build fails with "Python not found" RTMprofile probes for RTMcompare's bundled Python first. If RTMcompare isn't installed, falls back to system `python3` — which needs `numpy` , `scipy` , `soundfile` , `pyloudnorm` already installed. Install RTMcompare and the problem disappears.

Versions + spec dates

The app shows its version in the lower-right of the header. Each analysis stamps the spec versions it ran against. Reload an old result and the **Spec Drift** badge tells you whether anything has shifted.

This manual covers **5.0.5**.